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Fırat Diker[a)]

AFFILIATIONS

International Centre for Theory of Quantum Technologies, University of Gdansk, 80-309 Gdansk, Poland

[a)]Author to whom correspondence should be addressed: diker_firat@hotmail.com

ABSTRACT

We propose a quantum circuit composed of *cNOT* gates and four single-qubit gates to generate a *W* state of three qubits. This circuit was then enhanced by integrating two-qubit gates to create a *W* state of four and five qubits. After a couple of enhancements, we show that an arbitrary *W* state can be generated depending only on the degree of enhancement. The generalized formula for the number of two-qubit gates required is given, showing that an *n*-qubit *W*-state generation can be achieved with quadratically increasing number of two-qubit gates. The practical feasibility is discussed regarding photon sources and various applications of *cNOT* gates. In addition, we discuss possible future directions for quantum-state networks, concerning quantum correlations of *W* states.

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Quantum entanglement is a vital concept for understanding many quantum information and computational tasks. Since the famous EPR paper,[1] the entanglement of quantum states has attracted many scientists' attention. To understand this quantum phenomenon, fundamental features of entanglement have been studied.[2] In the field of quantum information processing, bipartite entanglement has been better understood by studying its creation, quantification, and manipulation.[2–6] Using local operations, bipartite entangled states can be converted from one to another, but it has been claimed that this is not possible for multipartite entangled states.[7] However, recent findings show that transforming one class of multipartite entangled states to another is achievable under stochastic local operations and classical communication (SLOCC). Entanglement transformation between *W* and *GHZ* states has been shown, and the conversion rate between these states has been studied using the concepts of degeneration and border rank of tensors from algebraic complexity theory.[8] In another work, it has been proven that obtaining a *W* state from a *GHZ* state with unit rate is possible, and transforming *GHZ* states into *W* states has also been shown.[9] These methods are important since they open alternative ways of building large-scale multipartite networks. *W*- and *GHZ*-state generation schemes can be compared regarding practical and experimental feasibility, and if the methods for the creation of the desired type are less feasible, one can prefer using a scheme to get the other type of state, which is to be transformed into the desired type via SLOCC as discussed in Refs. 8 and 9. Because multipartite entangled states are required to implement some quantum information tasks, it is necessary to construct large-scale quantum-state networks. For example, multipartite entangled states are used for quantum teleportation[10] and quantum key distribution.[11] In addition, some specific tasks require a particular type of multipartite entangled states, such as *GHZ* states,[12] for reaching a consensus in distributed networks, and *W* states are required for realizing an optimal universal quantum cloning machine.[13] *GHZ* and cluster states have been created,[14,15] but because of its sophisticated structure, *W* states are relatively hard to create in large scales. Some theoretical and experimental reports have proposed expanding a *W* state with ancillary photons or fusing two *W* states.[16–22] Özdemir *et al.*[23] succeeded in fusing two *W* states to generate large-scale *W* states. Large-scale *W* states are important to construct high capacity information processors. They proposed an optical setup that fused two *W* states of arbitrary sizes larger than or equal to 3. By integrating a Fredkin gate into this setup, a larger-scale resultant *W* state was obtained with a higher probability.[24] There are also theoretical

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proposals to fuse three and four W states[25,26] on optical setups, including the basic fusion gate.[23] As the size of the input W states increased, the success probability of fusion decreased, increasing the resource cost for these setups. An optical scheme has been proposed, showing that it is possible to expand W states deterministically.[27] This scheme doubles the size of an arbitrary W state deterministically by accessing locally all N qubits of the input W state. This work is very similar to our proposal regarding the primary purpose for generating large-scale W states deterministically; however, their expansion schemes can be used to obtain W states with an even number of polarization-encoded qubits. Although an arbitrary-size W state can be expanded by doubling its size, the resultant W state will always have an even number of qubits. One can detect one of the qubits to make a W state collapse into a smaller size by one qubit. This action makes the process probabilistic, defeating the main purpose of generating W states with unit probability. Moreover, if one wants to expand a W state with an odd number of qubits, we need to ask how one obtained the input W state, given that it is impossible to do so deterministically by using the same method. In the current work, we will be able to obtain an arbitrary-size W state deterministically, which means that W states with both odd and even numbers of qubits can be generated. There are also protocols proposed for concentrating arbitrary less-entangled W state into a maximally entangled W state.[28,29] Instead of expansion or concentration, we here perform creation operation to obtain a W state of arbitrary size, which is also a deterministic process. The current scheme is all-optical setup and requires only single- and two-qubit gates. The successful application probabilities of these gates, which are not considered at this point, will be discussed later.

In this work, we propose quantum circuits that can be implemented to generate a W state of any size using a certain number of non-entangled photons. First, we showed that three photons can be used as inputs to obtain a W state of three qubits. Then, we increased our number of input photons by one to create a W state of four qubits. After analyzing these two algorithms, we noticed that we could continue to enhance our W-state generating circuit to increase the size of the resultant W state. This led to the conclusion that we can use n number of photons to generate an n-qubit W state using only two-qubit gates. These gates are F gates, previously shown to work in an optical setup,[27] and $cNOT$ gates. The size of our resultant W state depends on how many photons are used as input qubits. The number of photons is equivalent to the number of qubits of the resultant W state. A $cNOT$ gate is a two-qubit gate that changes the polarization of the target qubit when the control qubit is vertically (V) polarized. The F gate is also a two-qubit gate composed of four half-wave plates (HWPs) and a $cNOT$ gate. The F gate acts on the target qubit when the control qubit is V polarized. The action of the F gate depends on how we arrange the HWPs on the circuit. Now, let us look at the case where three photons are put into our circuit shown in Fig. 1. There are two F gates and two $cNOT$ gates. The input photon in spatial mode 1 is V polarized, whereas the other photons are horizontally (H) polarized. In each ket, the left qubit is in spatial mode 1 and the number of spatial modes increases from left to right. The F gate in red transforms differently from the transformation of the F gate in black as shown below. The black F gate is a controlled Hadamard gate because it is composed of HWPs working at an angle of $\pi/8$. The F gate is a generalized term for all

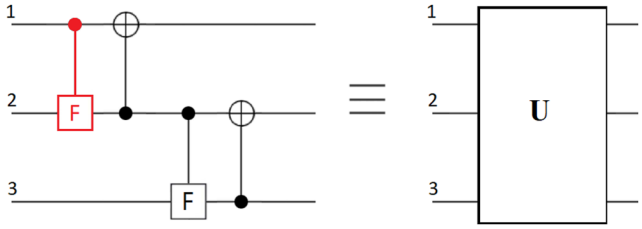


FIG. 1. Three photons are sent into the circuit consisting of two F gates and two $cNOT$ gates. All the two-qubit operations are shown in the U box for further use. The input state, $|VHH\rangle$, is transformed into a W state of three qubits.

angles. $cNOT$ gates are denoted by $cNOT$, and F gates are denoted by F. Because the two-qubit gates act on different subgroups of the input photons, we put two subindices showing the spatial modes of the control and target qubit, respectively. All transformations of the gates including the $cNOT$s are as follows:

$$\begin{aligned} F_{23}cNOT_{21}F_{12}|VHH\rangle &= F_{23}cNOT_{21} \\ &\quad\times\left[|V\rangle\otimes\left(\frac{1}{\sqrt{3}}|H\rangle+\sqrt{\frac{2}{3}}|V\rangle\right)\otimes|H\rangle\right] \\ &= F_{23}cNOT_{21}\left[\frac{1}{\sqrt{3}}|VHH\rangle+\sqrt{\frac{2}{3}}|VVH\rangle\right] \\ &= F_{23}\left[\frac{1}{\sqrt{3}}|VHH\rangle+\sqrt{\frac{2}{3}}|HVH\rangle\right] \\ &= \frac{1}{\sqrt{3}}(|VHH\rangle+|HVH\rangle+|HVV\rangle). \end{aligned} \tag{1}$$

Finally, the last $cNOT$ gate acts on our state, resulting in a W state of three qubits,

$$\begin{aligned} &cNOT_{32}\left[\frac{1}{\sqrt{3}}(|VHH\rangle+|HVH\rangle+|HVV\rangle)\right] \\ &\quad=\frac{1}{\sqrt{3}}(|VHH\rangle+|HVH\rangle+|HHV\rangle). \end{aligned} \tag{2}$$

When we use four two-qubit gates, we achieve the creation of a W state containing three qubits.

Now, let us look at the case where we use four photons to create a W state of four qubits. The circuit consists of three F gates and five $cNOT$s as shown in Fig. 2. This circuit is shown with the U box (Fig. 3). The action of the U box is as follows: $U|VHH\rangle = |W_3\rangle$, where $|W_3\rangle$ is a three-qubit W state. Let us show our operations step by step, as follows:

$$F_{12}|VHHH\rangle = \frac{1}{\sqrt{4}}|VHHH\rangle+\sqrt{\frac{3}{4}}|VVHH\rangle. \tag{3}$$

The U box acts only on the second component of the resultant state because the photon in the spatial mode 2 is $|V\rangle$, and therefore,

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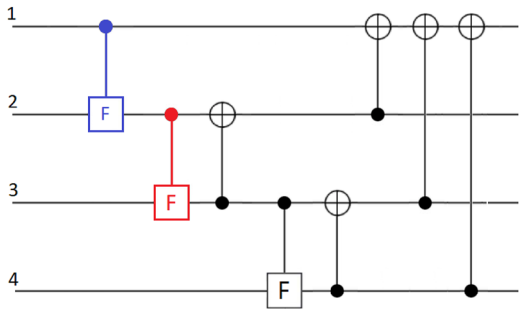


FIG. 2. Optical scheme, which is the enhanced version of the three-qubit *W*-state creation circuit. This circuit is composed of eight two-qubit gates, of which three are *F* gates. The other ones are *cNOT* gates. Four photons are sent into the circuit to create a *W* state of four qubits.

$$\begin{aligned} UF_{12}|VHHH\rangle &= U\left[|V\rangle \otimes \left(\frac{1}{\sqrt{4}}|HHH\rangle + \sqrt{\frac{3}{4}}|VHH\rangle\right)\right] \\ &= |V\rangle \otimes \left(\frac{1}{\sqrt{4}}|HHH\rangle + \sqrt{\frac{3}{4}}|W_3\rangle\right) \\ &= \frac{1}{\sqrt{4}}(|VHHH\rangle + |VVHH\rangle + |VHVH\rangle + |VHHV\rangle). \end{aligned} \tag{4}$$

The last three *cNOT* gates lead to a *W* state of four qubits, which is

$$\begin{aligned} &cNOT_{21}cNOT_{31}cNOT_{41} \\ &\quad \times \left[\frac{1}{\sqrt{4}}(|VHHH\rangle + |VVHH\rangle + |VHVH\rangle + |VHHV\rangle)\right] \\ &= \frac{1}{\sqrt{4}}(|VHHH\rangle + |HVHH\rangle + |HHVH\rangle + |HHHV\rangle) \\ &= |W_4\rangle. \end{aligned} \tag{5}$$

By adding an extra photon to the system again as shown in Fig. 4, we can obtain a five-qubit *W* state. The previous network, which is used to create a *W* state of four qubits, is shown as $\tilde{U}$. The input photons are transformed as follows:

$$\tilde{U}F_{12}|VHHHH\rangle = \frac{1}{\sqrt{5}}|VHHHH\rangle + \sqrt{\frac{4}{5}}|V\rangle \otimes |W_4\rangle, \tag{6}$$

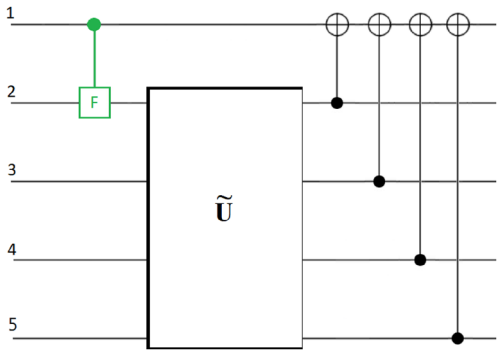


FIG. 4. Creation scheme for a five-qubit *W* state. This circuit is composed of four *cNOT* gates, a *F* gate, and $\tilde{U}$ shown in Fig. 3. The number of gates added to the previous scheme, $\tilde{U}$, is equal to the number of qubits belonging to the resultant state.

TABLE I. Number of gates used for the creation of certain *W* states.

Resultant *W* state	No. of two-qubit gates	No. of *F* gates	No. of *cNOT*s
$\|W_3\rangle$	4	2	2
$\|W_4\rangle$	8	3	5
$\|W_5\rangle$	13	4	9
$\|W_6\rangle$	19	5	14
$\|W_7\rangle$	26	6	20

$$\begin{aligned} &cNOT_{21}cNOT_{31}cNOT_{41}cNOT_{51}\tilde{U}F_{12}|VHHHH\rangle \\ &\quad = cNOT_{21}cNOT_{31}cNOT_{41}cNOT_{51} \\ &\qquad \times \left[\frac{1}{\sqrt{5}}|VHHHH\rangle + \sqrt{\frac{4}{5}}|V\rangle \otimes |W_4\rangle\right] \\ &\quad = \frac{1}{\sqrt{5}}(|VHHHH\rangle + |HVHHH\rangle + |HHVHH\rangle \\ &\qquad + |HHHVH\rangle + |HHHHV\rangle) \\ &\quad = |W_5\rangle. \end{aligned} \tag{7}$$

We have constructed three circuits using 4 two-qubit gates, 8 two-qubit gates, and 13 two-qubit gates to obtain three-, four- and five-qubit *W* states, respectively. When we want to increase the

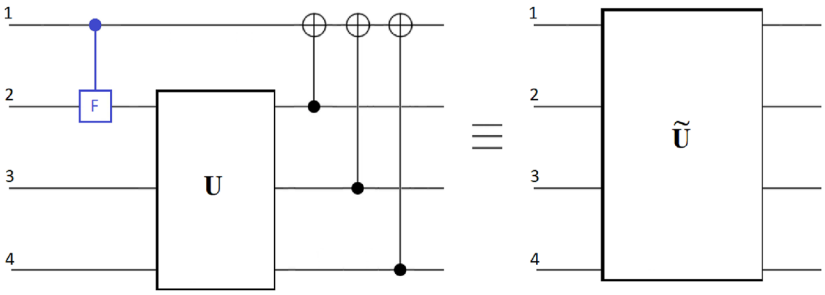


FIG. 3. Circuit used for creating a four-qubit *W* state. By integrating three *cNOT*s and one *F* gate, we enhance the circuit *U* that gives us a three-qubit *W* state. By enhancing the network, we are able to create a larger-scale *W* state by one qubit. The larger-scale *W* state is a four-qubit state. The whole network is shown as $\tilde{U}$.

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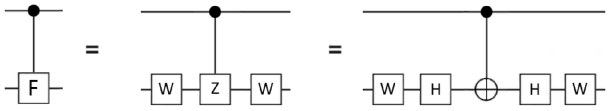


FIG. 5. Decomposition of the *F* gate into four *HWP*s and a *cNOT* gate. *W* gates are *HWP*s working at a certain angle that determines the transformation of a specific *F* gate. Hadamard gates can be implemented using *HWP*s working at the angle $\pi/8$. However, *HWP*s are not equivalent to a Hadamard gate because of its nonunitary nature. Therefore, we use *HWP*s to perform Hadamard operation.

number of qubits of our *W* state by one qubit, we integrate extra *n* two-qubit gates, where *n* is the size of the resultant *W* state. By performing straightforward calculations, the generalized formula for the number of gates is found as

$$\frac{n(n+1)-4}{2}, \tag{8}$$

where *n* is the size of the resultant *W* state. The number of *F* gates used is $n-1$, whereas the number of *cNOT* gates used is $\frac{(n-2)(n+1)}{2}$. To create a *W* state of *n* qubits, $n-1$ *F* gates are used, with transformations as follows:

$$\begin{aligned}
F_{(n-1)n}|10\rangle &= |1\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle),\\
F_{(n-2)(n-1)}|10\rangle &= |1\rangle \otimes \left(\frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle\right),\\
&\vdots\\
F_{23}|10\rangle &= |1\rangle \otimes \left(\frac{1}{\sqrt{n-1}}|0\rangle + \sqrt{\frac{n-2}{n-1}}|1\rangle\right),\\
F_{12}|10\rangle &= |1\rangle \otimes \left(\frac{1}{\sqrt{n}}|0\rangle + \sqrt{\frac{n-1}{n}}|1\rangle\right),
\end{aligned} \tag{9}$$

where the subindices show the spatial modes of the control and target qubits, respectively. One can also look at Table I to see how many gates we used for the creation of certain *W* states. The total number of two-qubit gates needed to obtain a *W* state of *n* qubits is proportional to n^2, which means that it increases quadratically.

Previously, we defined how *F* gates act. Each of these gates can be decomposed into three optical elements that are two *HWP*s working at some rotation angles, θ, and a *cZ* gate. In addition, a *cZ* gate can be decomposed into a *cNOT* gate and two *HWP*s acting as Hadamard gates. So, the total number of two-qubit gates needed is equivalent to the number of *cNOT*s to be used. In Fig. 5, the optical circuit corresponding to the *F* gate is shown, and their matrix representations are as follows:

$$W = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}, cZ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \tag{10}$$

Each transformation of the *F* gates depends on the angle at which the half-wave plates manipulate the polarization of the photons. The general formula of a *F* gate is

$$F = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos 2\theta & \sin 2\theta \\ 0 & 0 & \sin 2\theta & -\cos 2\theta \end{pmatrix}, \tag{11}$$

where the sine and cosine elements determine the factors of the components of the output state. A graph showing the relationship between the number of qubits *n* and the total number of *cNOT*s is shown in Fig. 6. The relationship between the size of the resultant *W* state and the angle of the first *HWP* is shown in Fig. 7. The mathematical relation is

$$\theta = \frac{1}{4}\arccos\left[\frac{1}{\sqrt{n}}\right]. \tag{12}$$

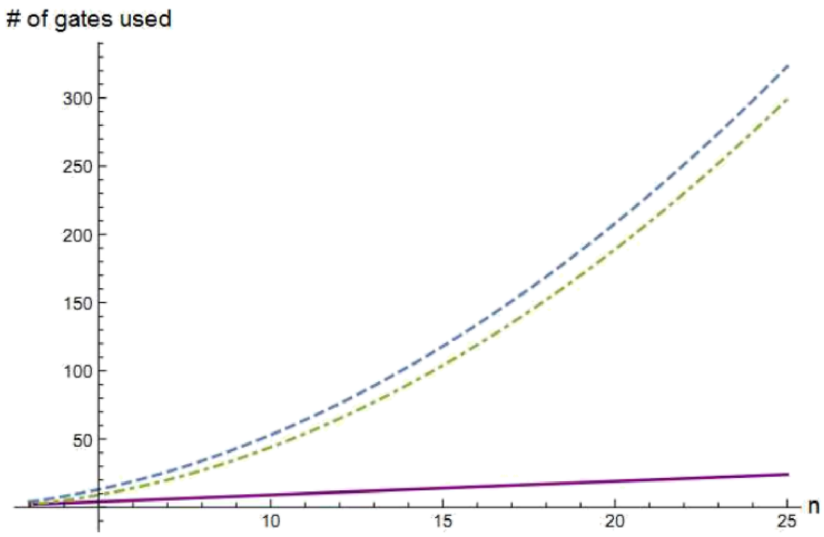


FIG. 6. Graph showing the relationship between the total number of two-qubit gates required and the number of qubits belonging to the resultant *W* state (blue dashed line), the relationship between the number of *cNOT*s and *n*, the number of qubits of *W* state (green dotted-dashed line), and the relationship between *n* and the number of *F* gates used (purple solid line).

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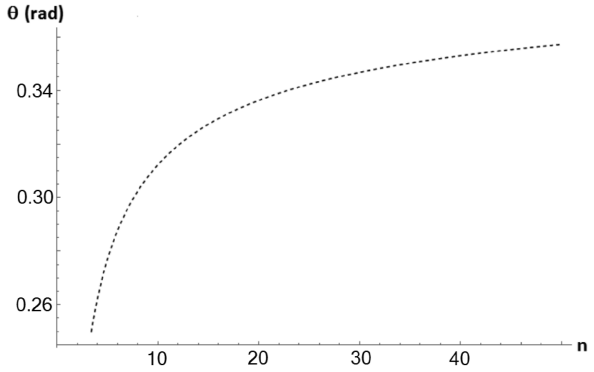


FIG. 7. A graph showing the relationship between the size of the resultant *W* state and the angle at which the first *HWP* acts. All the other *HWP*s act as shown in Eq. (9). The relationship between the angles of the other *HWP*s and *n* can also be seen in the graph. This graph shows how we should arrange our *HWP*s such that the circuit generates a *W* state of *n* qubits.

Although the proposed schemes allow us to create any desired W state, we lack ideal conditions regarding experimental realization for the sophisticated structure of these methods. One of the drawbacks of our setup is that as the size of a *W* state gets larger, the number of required gates increases quadratically. Therefore, this leads to a large number when we would like to create large-scale *W* states. For a large *n*, we have shown that the number of *cZ* gates is $\propto n$. This value is very small compared with the number of *cNOT* gates, that is, $\propto n^2$. Because the action of the *cZ* gate corresponds to the operation done by two *HWP*s and a *cNOT* gate, we need to consider the implementation of the *cNOT* gate. Here, we emphasize the importance of the experimental implementation of the *cNOT* gate by showing that large-scale *W* states can be created using only two-qubit gates. There are theoretical and experimental proposals for implementing the *cNOT* gate. In addition, the *cNOT* gate has been experimentally shown to work with a probability of 1/9 in linear optics.[30] This means that the probability of the successful creation of an *n*-qubit *W* state is $(1/9)^{\frac{n(n+1)-4}{2}}$. Successful creation probability is very low, even for a three-qubit *W* state with a success probability in the order of 10^{-4}. However, there have been proposals to implement optical gates using Kerr nonlinearities.[31–33] In one of these, Nemoto and Munro[33] pointed out that weak cross-Kerr nonlinearities are more practical to construct a *cNOT* gate with fewer physical resources than other linear optical schemes. They have shown that the almost deterministic *cNOT* gate can be realized in an optical circuit. By using the same method, the photonic *cNOT* gate can be realized with the success probability close to unit (with also high fidelity).[34] Therefore, our work increases the importance of cross-Kerr nonlinearities due to their application for *cNOT* gates. Through this method, our W-state-generation schemes are more feasible regarding the experimental realization of *cNOT* gates working with high success probability. In other words, it is crucial to have (almost) deterministic quantum gates. There are also other methods to realize a *cNOT* gate using superconductors[35] and ion-trap systems.[36,37]

Another experimental imperfection is that, if *n* is very large, the process may lead to failure case since a few degrees of deviation in angle changes the outcome, which means the desired *W* state is not created. For example, to create a *W* state of 100 qubits, we need to set the angle at 22.05° for the first *HWP*. If we accidently set the angle at 21.5°, we get a failure case because this angle is approximately the angle required for the first *HWP* to get a *W* state of 200 qubits. This small change in the angle changes the output greatly, leading to failure for the creation process. Theoretically, we here assume that the angle setting is adjusted as required for a *W*-state of *n* qubits. The precision of angles will be a challenging issue for experimentalists. In addition, the effect of photon loss (the amplitude damping channel) may lead to experimental imperfections as discussed in Ref. 38.

We should also consider the practical source of single photons to consider the feasibility of the proposed schemes. Parametric downconversion (PDC) is a common and well-known method to generate single-photon states and has been discussed for the scheme expanding polarization entangled *W* states.[17] Suppose that single photons are generated from PDC with rate γ. Three single photons required to prepare a three-qubit *W* state are generated with rate γ^3. In this case, the undesirable events usually occur due to generating an extra photon pair, with rate δ ($\sim 10^{-4}$). Such events occur with rate $O(\gamma^3\delta)$, which is very small compared with the rate $O(\gamma^3)$. It can be generalized to the case of n-photon *W*-state creation, which gives the rates of the desired events and errors, $O(\gamma^n)$ and $O(\gamma^n\delta)$, respectively. As in the simplest example of three-photon *W*-state creation, the rate of errors is small compared with the rate of the desired events. One can use quantum dots (QDs) in nanophotonic structures as a good source of single photons.[39–43] Along this direction, a planar nanophotonic circuit has been presented, where we can observe deterministic excitation of QDs.[44] In this work, the generated photons and the laser can be separated by using two orthogonal waveguide modes. The authors have shown a source emitting single photons with high indistinguishability and purity, and the coupling efficiency into the waveguide is >80.

We have presented three optical setups that deterministically create *W* states of three, four, and five qubits. Then, we have shown that a *W* state of any size can be created deterministically by enhancing the network. We have also derived the formula for the number of gates needed for this processing. The important capability of our proposal is that one can obtain any *W* state using a number of gates depending on the desired resultant *W* state size. In addition, we only need two-qubit gates, *cZ* and *cNOT*, making our setup more applicable because we do not use any three- or more-qubit gates. The implementation of the *cNOT* gate has been well characterized in the literature, shown in both theoretical and experimental studies. W states are known to have sophisticated structures due to entanglement among their qubits. The degree of entanglement is given by concurrence,[45] and the quantum non-locality is parameterized by this measure.[46,47] Another unique feature of quantum states, called contextuality, is also parameterized by concurrence.[48] That is to say, W states, as a specific class of quantum states, are non-local. Due to their sophisticated structures, the non-locality of W states is robust against noises, and its robustness has been investigated under particle loss.[49] When all but two particles are removed from a W state with n qubits, the resultant Bell state, including two qubits, is still entangled. The robustness of non-locality in W-state systems has also been investigated.[49,50] The non-locality of W states has been examined under the well-known CHSH test[51] as well.[52] W states

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are fully symmetric, and there is symmetry in every pair of qubits belonging to a W state. It does not matter which pair you choose due to complete symmetry. Intuitively, one may argue that W states are contextual as well because the symmetric two-qubit states are quantum-contextual.[48] As one can see, quantum correlations (non-locality and contextuality) are essential observables that give insights into quantum information processing. In the future, we should look into these correlations more for practical purposes and efficiency in information technologies.

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AUTHOR DECLARATIONS

Conflict of Interest

The author has no conflicts to disclose.

Author Contributions

Fırat Diker: Methodology (lead); Writing – original draft (lead).

DATA AVAILABILITY

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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